```sql
--What is the total revenue generated by male vs. female customers?
select gender, SUM(purchase_amount) as revenue
from "Customer"
group by gender
```

a Output Messages Notifications

| gender text | revenue numeric |
|---|---|
| Female | 75191 |
| Male | 157890 |

```sql
--Which customers used a discount but still spent more than the average purchase amount?
select customer_id, purchase_amount
from "Customer"
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from "Customer")
```

Data Output Messages Notifications

Showing rows: 1 to 839 Pag

| | customer_id bigint | purchase_amount bigint |
|---|---|---|
| 1 | 2 | 64 |
| 2 | 3 | 73 |
| 3 | 4 | 90 |
| 4 | 7 | 85 |
| 5 | 9 | 97 |
| 6 | 12 | 68 |
| 7 | 13 | 72 |
| 8 | 16 | 81 |
| 9 | 20 | 90 |

```sql
--Which are the top 5 products with the highest average review rating?
select item_purchased, round(avg(review_rating::numeric),2) as "Average Product Rating"
from "Customer"
group by item_purchased
order by avg(review_rating) desc
limit 5
```

Data Output Messages Notifications

Showing rows: 1 to 5

| item_purchas text | Average Product Rating numeric |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.80 |
| Skirt | 3.78 |

```sql
--Compare the average Purchase Amounts between Standard and Express Shipping.
select shipping_type,
ROUND(AVG(purchase_amount::numeric),2)
from "Customer"
where shipping_type in ('Standard','Express')
group by shipping_type;
```

| shipping_type text | round numeric |
|---|---|
| Standard | 58.46 |
| Express | 60.48 |

```sql
--Which 5 products have the highest percentage of purchases with discounts applied?
SELECT item_purchased,
       ROUND(100.0 * SUM( --when discount applied for a product in a group then count1 else 0
       CASE WHEN discount_applied = 'Yes'
       THEN 1 ELSE 0
       END
       )/COUNT(*),2) AS percentage_purchase
FROM "Customer"
GROUP BY item_purchased
ORDER BY percentage_purchase DESC
LIMIT 5;
```

Showing rows: 1 to 5

| item_purchased text | percentage_purchase numeric |
|---|---|
| Hat | 50.00 |
| Sneakers | 49.66 |
| Coat | 49.07 |
| Sweater | 48.17 |
| Pants | 47.37 |

```sql
--Segment customers into New, Returning, and Loyal based on their total
--number of previous purchases, and show the count of each segment.
with customer_type as ( --creating a cte called customer_type
SELECT customer_id, previous_purchases,
CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
    END AS customer_segment
FROM "Customer")
select customer_segment,count(*) AS "Number of Customers"
from customer_type
group by customer_segment;
```

| | customer_segment text | Number of Customers bigint |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | New | 83 |
| 3 | Returning | 701 |

```sql
--What are the top 3 most purchased products within each category?
WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
    FROM "Customer"
    GROUP BY category, item_purchased
)
SELECT item_rank,category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <=3;
```

Data Output Messages Notifications

Showing rows: 1 to 11 Page No: 1

| | item_rank bigint | category text | item_purchased text | total_orders bigint |
|---|---|---|---|---|
| 1 | 1 | Accessories | Jewelry | 171 |
| 2 | 2 | Accessories | Sunglasses | 161 |
| 3 | 3 | Accessories | Belt | 161 |
| 4 | 1 | Clothing | Blouse | 171 |
| 5 | 2 | Clothing | Pants | 171 |
| 6 | 3 | Clothing | Shirt | 169 |
| 7 | 1 | Footwear | Sandals | 160 |
| 8 | 2 | Footwear | Shoes | 150 |

```sql
--Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM "Customer"
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Data Output Messages Notifications

Showing rows: 1 to 2

| | subscription_status text | repeat_buyers bigint |
|---|---|---|
| 1 | No | 2518 |
| 2 | Yes | 958 |

```sql
--What is the revenue contribution of each age group?
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue
FROM "Customer"
GROUP BY age_group
ORDER BY total_revenue desc;
```

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| | age_group text | total_revenue numeric |
|---|---|---|
| 1 | Young Adult | 62143 |
| 2 | Middle Aged | 59197 |
| 3 | Adult | 55978 |
| 4 | Senior | 55763 |

```sql
--Do subscribed customers spend more? Compare average spend and total revenue
--between subscribers and non-subscribers.
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount),2) AS avg_spend,
       ROUND(SUM(purchase_amount),2) AS total_revenue
FROM "Customer"
GROUP BY subscription_status
ORDER BY total_revenue,avg_spend DESC;
```

| subscription_status text | total_customers bigint | avg_spend numeric | total_revenue numeric |
|---|---|---|---|
| Yes | 1053 | 59.49 | 62645.00 |
| No | 2847 | 59.87 | 170436.00 |